

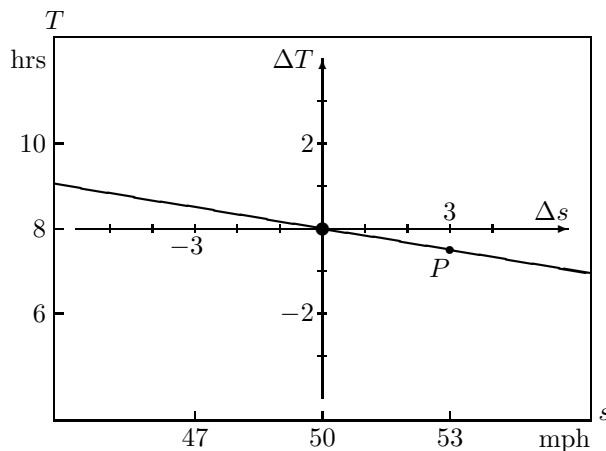
50 mph, travel time is cut by about $9\frac{1}{2}$ minutes. If we double the increase in speed, that doubles the savings in time: if the speed is 2 mph over 50 mph, travel time is cut by about 19 minutes. Compare this with a value of about 18 minutes that we got with the exact equation $T = 400/s$.

Δs and ΔT now have a special meaning

Notice that we are using Δs and ΔT in a slightly more restricted way than we have previously. Up to now, Δs measured the horizontal distance between *any* two points on a graph. Now, however, Δs just measures the horizontal distance from the fixed point $(s, T) = (50, 8)$ (marked with a large dot) that sits at the center of the window. Likewise, ΔT just measures the vertical distance from this point. The central point therefore plays the role of an origin, and Δs and ΔT are the *coordinates* of a point measured from that origin. To underscore the fact that Δs and ΔT are really coordinates, we have added a Δs -axis and a ΔT -axis in the window below. Notice that these coordinate axes have their own labels and scales.

Δs and ΔT are coordinates in the window

Every point in the window can therefore be described in two different coordinate systems. The two different sets of coordinates of the point labelled P , for instance, are $(s, T) = (53, 7.52)$ and $(\Delta s, \Delta T) = (3, -.48)$. The first pair says “When your speed is 53 mph, the trip will take 7.52 hrs.” The second pair says “When you increase your speed by 3 mph, you will decrease travel time by .48 hrs.” Each statement can be translated into the other, but each statement has its own point of reference.



The microscope equation: $\Delta T \approx -.16 \Delta s$ hours

We call $\Delta T \approx -.16 \Delta s$ the **microscope equation** because it tells us how the microscope coordinates Δs and ΔT are related.

In fact, we now have two different ways to describe how travel time is related to speed. They can be compared in the following table.

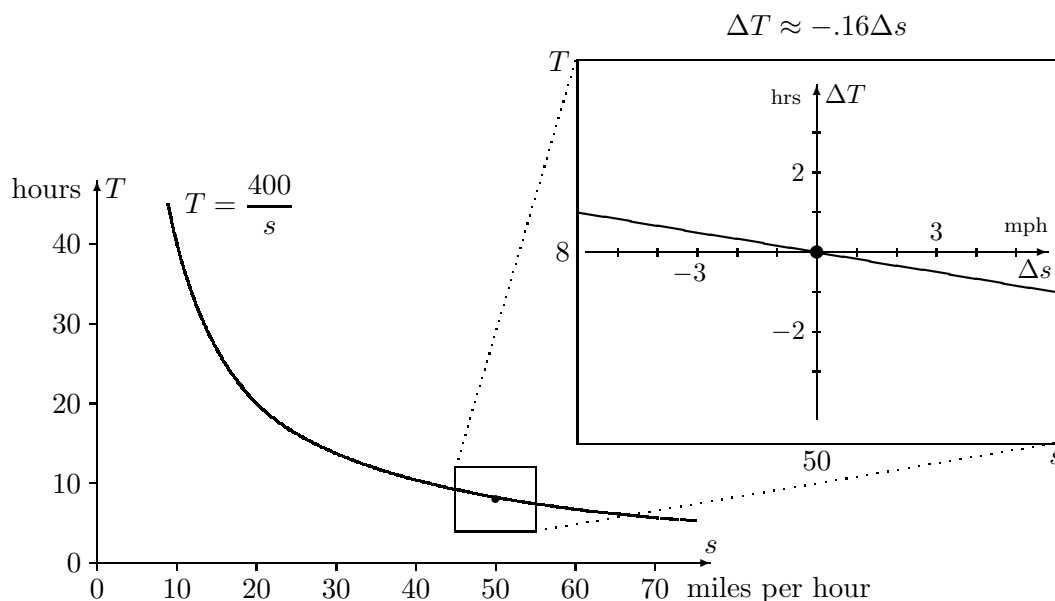
	GLOBAL	LOCAL
coordinates:	s, T	$\Delta s, \Delta T$
equation:	$T = 400/s$	$\Delta T \approx -.16 \Delta s$
properties:	exact non-linear	approximate linear

We say the microscope equation is *local* because it is intended to deal only with speeds near 50 miles per hour. There is a different microscope equation for speeds near 40 miles per hour, for instance. By contrast, the original equation is *global*, because it works for all speeds. While the global equation is exact it is also non-linear; this can make it more difficult to compute. The microscope equation is approximate but linear; it is easy to compute. It is also easy to put into words:

Global vs. local
descriptions

At 50 miles per hour, the travel time of a 400 mile journey decreases $9\frac{1}{2}$ minutes for each mile per hour increase in speed.

The connection between the global equation and the microscope equation is shown in the following illustration.



Local Linearity and Multipliers

The reasoning that led us to a microscope equation for travel time can be applied to any locally linear function. If $y = f(x)$ is locally linear, then at $x = a$ we can write

The microscope equation: $\Delta y \approx f'(a) \cdot \Delta x$

We know an equation of the form $\Delta y = m \cdot \Delta x$ tells us that y is a linear function of x , in which m plays the role of slope, rate, and multiplier. The microscope equation therefore tells us that **y is a linear function of x when x is near a** —at least approximately. In this almost-linear relation, the derivative $f'(a)$ plays the role of slope, rate, and multiplier.

The microscope equation is the analytic form of local linearity

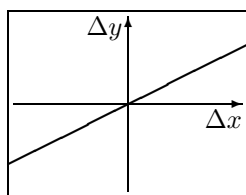
The microscope equation is just the idea of local linearity expressed analytically rather than geometrically—that is, by a formula rather than by a picture. Here is a chart that shows how the two descriptions of local linearity fit together.

$y = f(x)$ is locally linear at $x = a$:

GEOMETRICALLY

When magnified at $(a, f(a))$,
the graph of f is almost straight,
and the slope of the line is $f'(a)$.

microscope window



ANALYTICALLY

When x is near a ,
 y is almost a linear function of x ,
and the multiplier is $f'(a)$.

microscope equation

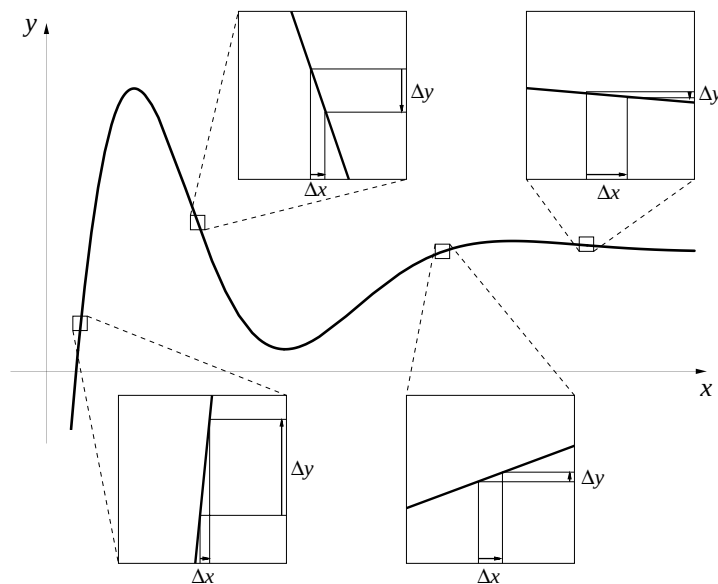
$$\Delta y \approx f'(a) \cdot \Delta x$$

Of course, the graph in a microscope window is not *quite* straight. The analytic counterpart of this statement is that the microscope equation is not *quite* exact—the two sides of the equation are only approximately equal. We write “ $\approx$ ” instead of “ $=$ ”. However, we can make the graph look even straighter by increasing the magnification—or, what is the same thing, by decreasing the field of view. Analytically, this increases the exactness of the microscope equation. Like a laboratory microscope, our microscope is most

accurate at the center of the field of view, with increasing aberration toward the periphery!

In the microscope equation $\Delta y \approx f'(a) \cdot \Delta x$, **the derivative is the multiplier that tells how y responds to changes in x** . In particular, a small increase in x produces a change in y that depends on the sign and magnitude of $f'(a)$ in the following way:

- $f'(a)$ is large and positive $\Rightarrow$ large increase in y ,
- $f'(a)$ is small and positive $\Rightarrow$ small increase in y ,
- $f'(a)$ is large and negative $\Rightarrow$ large decrease in y ,
- $f'(a)$ is small and negative $\Rightarrow$ small decrease in y .



For example, suppose we are told the value of the derivative is 2. Then any small change in x induces a change in y approximately twice as large. If, instead, the derivative is $-1/5$, then a small change in x produces a change in y only one fifth as large, and in the opposite direction. That is, if x increases, then y decreases, and vice-versa.

The microscope equation should look familiar to you. It has been with us from the beginning of the course. Our “recipe” $\Delta S = S' \cdot \Delta t$ for predicting future values of S in the S - I - R model is just the microscope equation for the function $S(t)$. (Although we wrote it with an “=” instead of an “ $\approx$ ” in the

The microscope equation is the recipe for building solutions to rate equations